DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 66

Question description

A company has n employees, who form a tree hierarchy where each employee has a boss, except for the general director.
Your task is to process q queries of the form: who is employee x's boss k levels higher up in the hierarchy?

Input

The first input line has two integers n and q: the number of employees and queries. The employees are numbered 1,2,…,n, and employee 1 is the general director.
The next line has n−1 integers e2,e3,…,en: for each employee 2,3,…,n their boss.
Finally, there are q lines describing the queries. Each line has two integers x and k: who is employee x's boss k levels higher up?

Output
Print the answer for each query. If such a boss does not exist, print −1.

Constraints

- 1≤n,q≤2·10^5
- 1≤ei≤i−1
- 1≤x≤n
- 1≤k≤n

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 3
1 3 2 2
4 2
1 2
1 3
```

EXPECTED OUTPUT

```
1
−1
−1
```

Test Case 2

INPUT (STDIN)

```
5 3
1 1 3 3
4 1
4 2
4 3
```

EXPECTED OUTPUT

```
3
1
−1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void link(int i,int j)
```

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
2	420	50

Code

solution.c

```c
#include <stdio.h>

int up[200005][20];

void link(int i, int j) {
    up[i][0] = j;
}

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) return 0;
    up[1][0] = 0;
    for (int i = 2; i <= n; i++) {
        int boss; scanf("%d", &boss);
        up[i][0] = boss;
    }
    for (int j = 1; j < 20; j++) {
        for (int i = 1; i <= n; i++) {
            if (up[i][j - 1] != 0) up[i][j] = up[up[i][j - 1]][j - 1];
            else up[i][j] = 0;
        }
    }
    while (q--) {
        int x, k;
        scanf("%d %d", &x, &k);
        for (int j = 19; j >= 0; j--) {
            if (k & (1 << j)) {
                x = up[x][j];
                if (x == 0) break;
            }
        }
        printf("%d\n", (x == 0) ? -1 : x);
    }
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
5 3
1 1 3 3
4 1
4 2
4 3
3
1
-1
PS C:\C Programming>
```